

# ARRAYS 3

Don't fear failure  
be afraid of not taking a  
chance

Car Story 3  
@ELLECOOPER.CO

Good  
Morning



## Topics

01. Merge Interval
02. Insert Interval
03. First Missing Integer

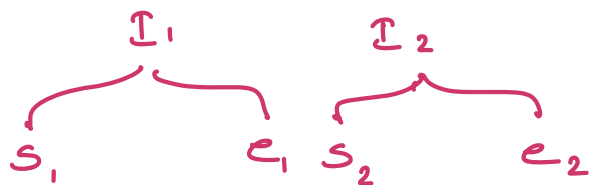
Starting at 7:06 AM

# Merged Interval

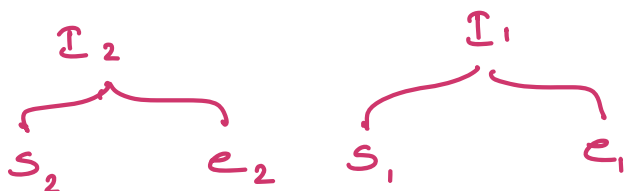
An interval is defined by start & end time.

| $I_1$    | $I_2$     |  | Merged Interval |
|----------|-----------|--|-----------------|
| $[2, 6]$ | $[3, 7]$  |  | $\{2, 7\}$      |
| $[2, 8]$ | $[4, 6]$  |  | $\{2, 8\}$      |
| $[2, 4]$ | $[5, 7]$  |  | No overlapping  |
| $[3, 7]$ | $[4, 10]$ |  | $\{3, 10\}$     |
| $[3, 6]$ | $[6, 10]$ |  | $\{3, 10\}$     |
| $[2, 5]$ | $[8, 10]$ |  | No overlapping  |
| $[5, 8]$ | $[1, 3]$  |  | No overlapping  |
| $[3, 9]$ | $[1, 4]$  |  | $\{1, 9\}$      |

\* Non overlapping

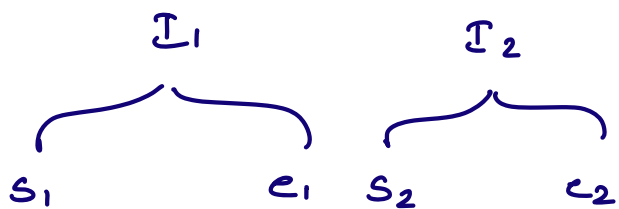


$$\Rightarrow e_1 < s_2$$



$$\Rightarrow e_2 < s_1$$

## \* Overlapping scenario



Merged Interval

$$\text{ans} = \{ \min(s_1, s_2), \max(e_1, e_2) \}$$

## \* Merge Intervals

Q Given a sorted list of overlapping intervals, sorted based on start time, merge all overlapping intervals & return sorted list.

$$\text{Interval}[] = \{ [0-2] [1-4] [5,6] [6,8] [7,10] [8,9] [12,14] \}$$

$\uparrow$  start       $\uparrow$  end  
 (under [0-2] and [6,8])

| Interval 1 | Interval 2 |                 | Ans                  |
|------------|------------|-----------------|----------------------|
| [0-2]      | [1-4]      | Overlapping     | [0-4]                |
| [0-4]      | [5-6]      | Non-overlapping | [0-4] [5,6]          |
| [5-6]      | [6-8]      | Overlapping     | [0-4] [5-8]          |
| [5-8]      | [7-10]     | Overlapping     | [0-4] [5-10]         |
| [5-10]     | [8-9]      | Overlapping     | [0-4] [5-10]         |
| [5-10]     | [12-14]    | Non-overlapping | [0-4] [5-10] [12-14] |

- \* Overlapping  $\rightarrow$  Merge interval & update last inserted one
- \* Non-overlapping  $\rightarrow$  Insert new interval

list < Interval > ans;

int st = A[0].start

int end = A[0].end

for (i=1; i < A.length; i++) {

if (A[i].start  $\leq$  end) { //merge overlapping

st = min(st, A[i].start);

end = max(end, A[i].end);

}

else {

ans.add(new Interval(st, end));

st = A[i].start;

end = A[i].end;

}

}

ans.add(new Interval(st, end));

TC:  $O(n)$

SC:  $O(1)$

Interval [ ] = { (0, 2) (2, 6) (5, 8) (9, 10) (9, 12) (11, 17) }

ans = { (0, 8) (9, 17) }

st = ~~0~~ 9

end = ~~2 6 8 10 12~~ 17

### \* Insert Interval

Q Given N non-overlapping intervals & they are sorted based on start time. You are given a new interval Merge the new interval & make all the non overlapping intervals

Eg: N = 7

New interval

New intervals after insertion

[1 3]

{1, 3}

[4 7]

{4 7}

[10 14] + {10 22} → {10, 22}

[16 19] + {10, 22} → {10, 22}

[21 24] + {10, 22} → {10, 24}

{10, 24}

{27, 30}

{32, 35}

[27 30]

{10, 24}

[32 35]

## \* Example 2

New Interval

[1 5]

[8 10]

[11 14] + {12, 22} → {11, 22}

[15 20] + {11, 22} → {11, 22}

[20 24] + {11, 22} → {11, 24}

Ans of non-overlapping  
intervals

{1, 5}

{8, 10}

{11, 24}

list <Interval> ans;

Ⓡ

for (i=0; i<n; i++){

if (ar[i].end < Ⓡ.start){

// Non overlapping

ans.add(ar[i]);

}

else if (ar[i].start > Ⓡ.end){ // Non-overlapping

ans.add(Ⓡ);

for (j=i; j<n; j++){

ans.add(ar[j]);

}

return ans;

}

else {

    I.start = min(ar[i].start, I.start);

    I.end = max(ar[i].end, I.end);

}

}

ans.insert(I);

return ans;

8:08 AM → 8:18 AM

\*